Table 2: Comparison of N_{GE} for different transfer learning methods across homogeneous and heterogeneous settings.

Source Device	**Target Device**	Direct Profiling (N^{DP}_{GE})	Direct Transfer (N^{DT}_{GE})	**Unsupervised Transfer Learning Attacks** (N^{US}_{GE})			
Homogeneous (Cross Device) N_{GE}							
				CDPA* [CZLG21]	ALPA* [CZG+22]	Dual-Leak$ [YWS+23]	**UTLA (Ours)**
$XMEGA_{pow}$	$XMEGA_{pow}$	12	357	15	17	16	**16**
$XMEGA_{EM}$	$XMEGA_{EM}$	14	737	58	57	55	**17**
SAKURA-AES	SAKURA-AES	586	1640	1425	1356	1524	**716**
Heterogeneous (Cross Domain) N_{GE}							
SAKURA-AES	$XMEGA_{pow}$	12	N.C.	3210	4123	N.C.	**61**
SAKURA-AES	$XMEGA_{EM}$	14	N.C.	4474	4895	N.C.	**142**
$XMEGA_{EM}$	$XMEGA_{pow}$	12	N.C.	1870	2980	3285	**47**
ASCADv1 (Random)	ASCADv1 (Fixed)	250	N.C.	N.C.	N.C.	N.C.	**1330**
ASCADv2	ASCADv1 (Fixed)	250	N.C.	N.C.	N.C.	N.C.	**1519**
Cache Side Channel	$XMEGA_{pow}$	195	N.C.	N.C.	N.C.	N.C.	**1233**
$XMEGA_{pow}$	Cache Side Channel	187	N.C.	N.C.	N.C.	N.C.	**435**

Remarks – N.C.: Training did not converge to GE = 1, which means that the attack cannot find the target key.; *: Attacks adapted from the codebase provided in the paper; $: In-house adaptation of the active learning algorithm.